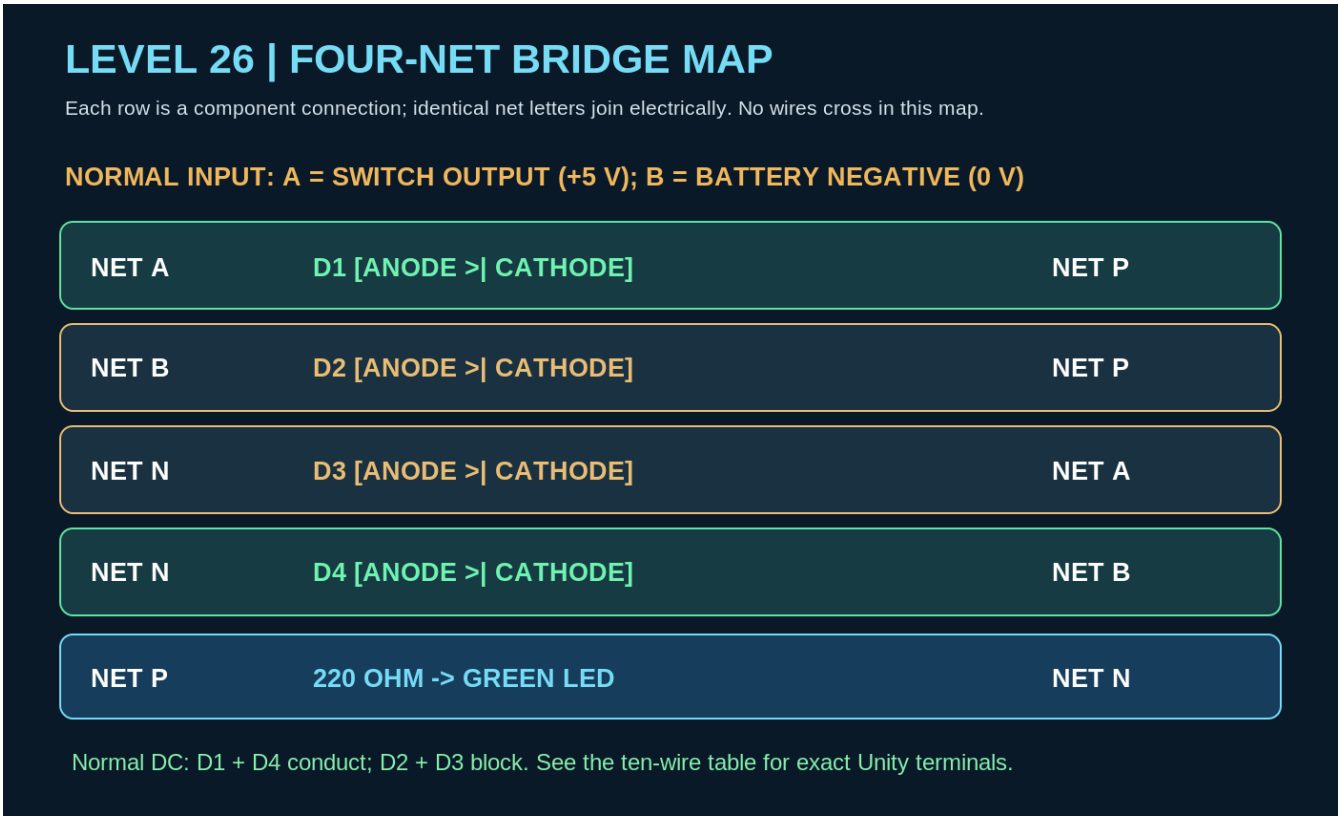
# PROJECT SPARK - LEVEL 26

## Bridge Defender - Four Diode Rectifier | Unity Creation Process

5 V DC | 4 x 1N4007 | 220 ohm | green LED | 10 wires | 6 objectives




## Complete terminal-to-terminal wiring

| | | |
|---|---|---|
| 1 | Battery_positive | SwitchInput |
| 2 | SwitchOutput | D_BRIDGE_1_Anode |
| 3 | SwitchOutput | D_BRIDGE_3_Cathode |
| 4 | Battery_negative | D_BRIDGE_2_Anode |
| 5 | Battery_negative | D_BRIDGE_4_Cathode |
| 6 | D_BRIDGE_1_Cathode | R_LIMIT_Input |
| 7 | D_BRIDGE_2_Cathode | R_LIMIT_Input |
| 8 | R_LIMIT_Output | LED_Anode |
| 9 | LED_Cathode | D_BRIDGE_3_Anode |
| 10 | LED_Cathode | D_BRIDGE_4_Anode |

# Unity setup / implementation

## 1. Duplicate

Duplicate the working Level 25 prefab as Level_26_Layout, keep the source, switch and LED, add two diode workstations, and change the resistor to 220 ohm.

## 2. Configure Smart Slots

Five unique slots: Slot_01 resistor plus Slot_02 to Slot_05 diodes. Enable Smart Binding for each. Set bridge Use Smart Slots = ON.

## 3. Declare runtime IDs

In the original Level 26 prefab Scene Bindings, declare all five component IDs and ten terminal IDs listed in the binding table. Preserve the fixed LED and source IDs.

## 4. Connect wires

Use the ten-wire table on page 1. Shared nodes need multiple wire connections per terminal. D1 and D4 conduct at normal 5 V polarity.

## 5. Import definition

Select the original Level_26.asset in Unity and import Level_26_Bridge_Rectifier_FIXED.txt; the original GUID is preserved.

## 6. Generate drawing asset

Put SparkLevel26DrawingAssetCreator.cs in Assets/Editor and select Tools > Project Spark > Visual Teaching > Create Level 26 Drawing Asset.

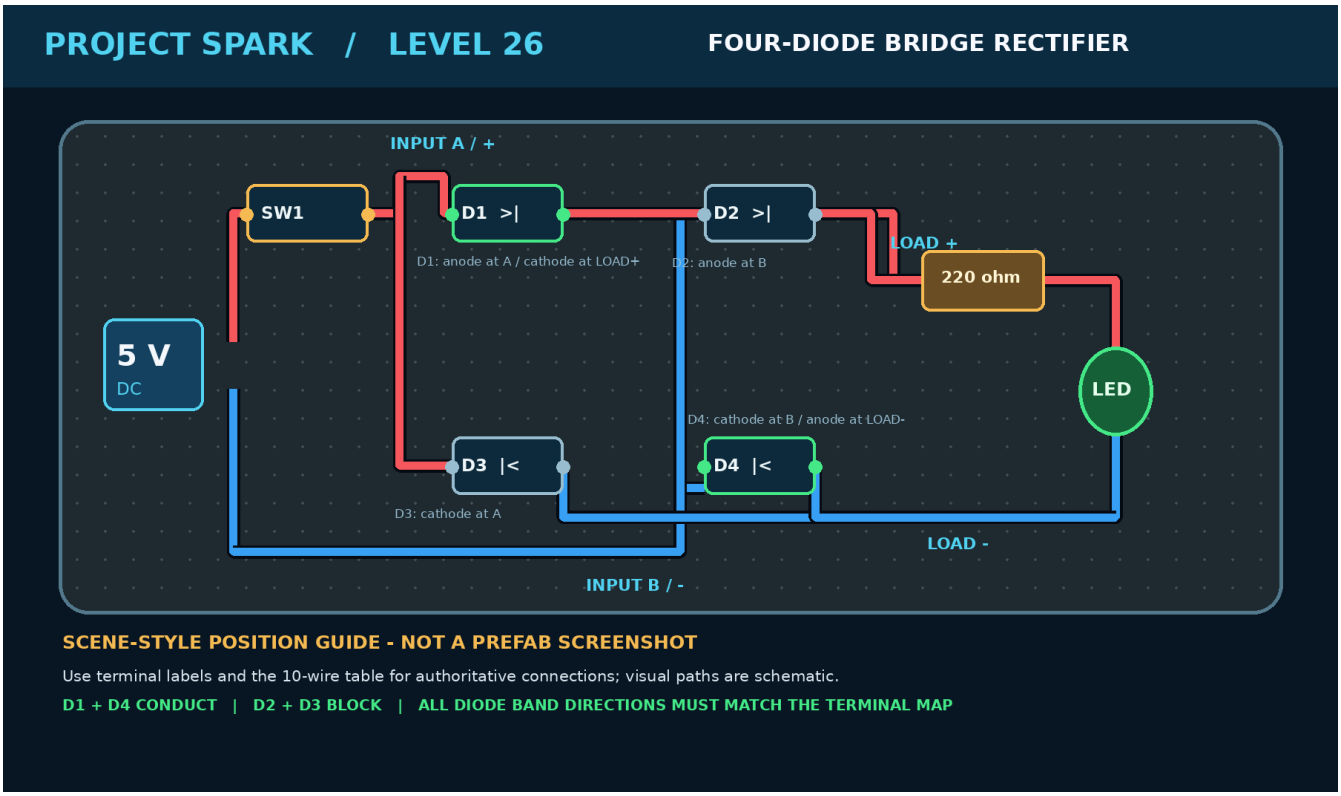
## 7. Test

Verify all 15 runtime IDs bound, LED illuminated, source 5 V, D1/D4 0.45-1 V, resistor 1.3-1.9 V, current 6-9 mA, 6/6 targets.

## Smart Slot ID map

| Slot | Component ID | Terminal A / Terminal B |
|---|---|---|
| Slot_01 | R_LIMIT_Component | R_LIMIT_Input / R_LIMIT_Output |
| Slot_02 | D_BRIDGE_1_Component | D_BRIDGE_1_Anode / D_BRIDGE_1_Cathode |
| Slot_03 | D_BRIDGE_2_Component | D_BRIDGE_2_Anode / D_BRIDGE_2_Cathode |
| Slot_04 | D_BRIDGE_3_Component | D_BRIDGE_3_Anode / D_BRIDGE_3_Cathode |
| Slot_05 | D_BRIDGE_4_Component | D_BRIDGE_4_Anode / D_BRIDGE_4_Cathode |

# Level 26 - Scene-style placement guide




## Expected results and verification

- D1, D4: forward, approximately 0.7 V each, current ~7.3 mA
- D2, D3: reverse-biased, approximately 0 A
- LED: illuminated, approximately 2.0 V
- 220 ohm resistor: about 1.6 V, current 6-9 mA
- Source: 4.8-5.2 V; closed return YES; shorts NO
- All six objectives verified via LevelGamePlayManager and current audit

These are nominal model estimates. The scene-style visual is schematic and not a literal screenshot; the terminal wiring table is authoritative.

## Important limitation

The base level checks only normal DC supply polarity. It does not automatically validate reversed input polarity or swapped conducting pairs. Do not alter the shared electrical solver just to create the level.